Alexander G. Lucaci, Ph.D.

Computational Biology — Bioinformatics — Viral Evolution — Urban Microbiome — Genomic Surveillance
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Computational biologist and bioinformatics leader developing scalable genomic analysis systems for viral discovery, molecular evolution, urban microbiome research, and public-health surveillance across large-scale sequencing datasets.

Leadership and Research Profile

- **Co-Chair, Bioinformatics Working Group, NIH Human Virome Program** **2025–Present**
  Set bioinformatics strategy for standardized viromics analysis, viral discovery, host profiling, data integration, public submission, and computational infrastructure for large-scale human virome studies.
- **Executive Director / Director of Bioinformatics, MetaSUB International Consortium** **2024–Present**
  Lead global urban microbiome and virome workflows across 100+ cities, emphasizing scalable pipelines, cross-site harmonization, public-health surveillance, and comparative microbial ecology.
- **Postdoctoral Associate, Systems and Computational Biomedicine, Weill Cornell Medicine** **2023–Present**
  Develop computational frameworks for evolutionary genomics, viromics, metagenomics, wastewater surveillance, and city-scale biological monitoring. Analyze large-scale genomic and metatranscriptomic datasets for NIH Human Virome Program and MetaSUB projects, including viral discovery, host profiling, taxonomic classification, AMR analysis, and metadata integration.
- **Graduate Research Assistant, Temple University** **2020–2023**
  Designed HyPhy-based molecular evolution methods to quantify multinucleotide substitutions and improve inference of episodic and gene-wide natural selection. Built comparative genomics workflows for dN/dS estimation, selection analysis, and scalable evolutionary inference across molecular sequence datasets.
- **Research focus:** Develop methods and workflows that connect molecular evolution, viral genomics, metagenomics, and environmental surveillance, with a focus on detecting evolutionary signals, building scalable analysis systems, and translating large-scale sequencing data into interpretable biological and public-health insights.

Selected Achievements

- **Scientific leadership:** Bioinformatics lead across major virome, urban microbiome, and genomic surveillance initiatives, including NIH HVP and MetaSUB.
- **Software ownership:** Lead developer of **RASCL** and **AOC**; contributor to HyPhy-based workflows for robust dN/dS and selection inference.
- **Funding and infrastructure:** NIH Loan Repayment Program Award, National Cancer Institute, **$100,000**; lead analyst and architect for NIH HVP U54 Data Analysis & Submission Core.
- **Research impact:** First-author publications spanning multinucleotide substitution models, rapid selection analysis workflows, and comparative evolutionary genomics.

Selected Publications

31 total publications, with first-author contributions in molecular evolution, selection analysis, comparative genomics, and reproducible bioinformatics software.

- **Lucaci AG**, Zehr JD, Enard D, Thornton JW, Kosakovsky Pond SL. Evolutionary shortcuts via multinucleotide substitutions and their impact on natural selection analyses. *Molecular Biology and Evolution*. 2023.
- **Lucaci AG**, Zehr JD, Shank SD, Bouvier D, Ostrovsky A, Mei H, Nekrutenko A, Martin DP, Kosakovsky Pond SL. RASCL: Rapid Assessment of Selection in CLades through molecular sequence analysis. *PLOS ONE*. 2022.
- **Lucaci AG**, Brew WE, Lamanna J, Selberg A, Carnevale V, Moore AR, Kosakovsky Pond SL. Evolution of mammalian Rem2 and implications for human disorders. *Frontiers in Bioinformatics*. 2024.

Software, Workflows, and Technical Skills

- **RASCL:** Reproducible workflow for rapid molecular selection analysis across viral and comparative genomic datasets.
- **AOC:** Workflow for characterizing natural selection across genes, orthologs, taxa, and biological conditions.
- **HVP and MetaSUB pipelines:** Viromics, viral discovery, host profiling, taxonomic profiling, AMR, metadata harmonization, public data submission, and scalable surveillance reporting.
- **AI-enabled genomic analysis:** Develop machine learning and AI-assisted approaches for literature mining, sequence interpretation, signal detection, workflow automation, and large-scale biomedical data analysis.
- **Technical strengths:** Reproducible workflow engineering, molecular evolution, phylogenetics, viromics, metagenomics, AMR profiling, scalable HPC analysis, Python/R development, and CI/CD-enabled research software.

Education, Awards, Teaching, and Service

Ph.D. Bioinformatics, Temple University, 2023; Dissertation: *The Role of Complex Evolutionary Dynamics in Molecular Sequence Analysis.* **M.S. Biology**, New York University, 2018. **B.S. Biochemistry**, Stony Brook University, 2011.

- **Awards:** NIH LRP Award, NCI, **$100,000**; SMBE Young Investigators Travel Award; Center for Open Science Opening Influenza Research Fellowship; Temple CST Outstanding Research Award.
- **Teaching, mentoring, and service:** Guest Lecturer, Weill Cornell Medicine; Instructor at international summer workshops, VEME (viral evolution) and EEBG (bioinformatics); Co-Founder, Temple Bioinformatics Studio; reviewer for *Astrobiology*, *Bioinformatics Advances*, *Molecular Biology and Evolution*, *Virus Evolution*, and *Genomics*.